Definition 1.26. A *number field* K is a finite extension of $\mathbb{Q}$. The *ring of integers* $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K.

Remark 1.27. The notation $\mathbb{Z}_K$ is also sometimes used to denote the ring of integers of K. The symbol $\mathcal{O}$ emphasizes the fact that $\mathcal{O}_K$ is an *order* in K; in any $\mathbb{Q}$-algebra K of finite dimension r, an order is a subring of K that is also a free $\mathbb{Z}$-module of rank r, equivalently, a $\mathbb{Z}$-lattice in K that is also a ring. In fact, $\mathcal{O}_K$ is the *maximal order* of K: it contains every order in K.

Proposition 1.28. *Let A be an integrally closed domain with fraction field K. Let α be an element of a finite extension L/K, and let $f \in K[x]$ be its minimal polynomial over K. Then α is integral over A if and only if $f \in A[x]$.*

Proof. The reverse implication is immediate: if $f \in A[x]$ then certainly α is integral over A. For the forward implication, suppose α is integral over A and let $g \in A[x]$ be a monic polynomial for which $g(\alpha) = 0$. In $\overline{K}[x]$ we may factor $f(x)$ as

$$f(x) = \prod_i (x - \alpha_i).$$

For each α_i we have a field embedding $K(\alpha) \to \overline{K}$ that sends α to α_i and fixes K. As elements of $\overline{K}$ we have $g(\alpha_i) = 0$ (since $f(\alpha_i) = 0$ and f must divide g), so each $\alpha_i \in \overline{K}$ is integral over A and lies in the integral closure $\tilde{A}$ of A in $\overline{K}$. Each coefficient of $f \in K[x]$ can be expressed as a sum of products of the α_i, and is therefore an element of the ring $\tilde{A}$ that also lies in K. But $A = \tilde{A} \cap K$, since A is integrally closed in its fraction field K. $\square$

Example 1.29. We saw in Example 1.24 that $(1 + \sqrt{5})/2$ is integral over $\mathbb{Z}$. Now consider $\alpha = (1 + \sqrt{7})/2$. Its minimal polynomial $x^2 - x - 3/2 \notin \mathbb{Z}[x]$, so α is not integral over $\mathbb{Z}$.

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